

Mechanical black box 2.0 — Solution

Y26 (2026) — English translation

01

-20.00 pts

Do not try to open the MCH! For opening the ball, you will be disqualified and the result of the round will be cancelled!

02

-1.00 pts

Carefully check the reliability of your installation, especially its moving parts! Repairing the MCH can last up to 30 minutes! Starting from the second, the breakdown of the MCH leads to a penalty of 1 point!

Starting from the second, the breakdown of the MCH leads to a penalty of 1 point!

A0

0.10 pts

Write down the number of the mechanical black box issued to you.

A1

0.10 pts

Measure and record the values of the mass $(M + m)$ and length L of the MCH.

Answer:

$$(M + m) = 199,2 \text{ g}$$

$$L = 45,0 \text{ cm}$$

A2

0.90 pts

Using drawings, describe the measurement method that allows you to obtain the product $m \cdot x_0$ of the mass of the bob and its equilibrium coordinate.

We will find the coordinate of the center of mass by placing the MCN on the edge of the table so that the MCN almost turns over. Let's take 2 measurements. In the first case, we smoothly transfer the MCH from a vertical position with a weight from below to a horizontal one, in the second, from a vertical position with a weight on top. This is necessary to average the static friction force.

Note. An alternative way to eliminate friction is to tap or roll the box on the table to bring the weight into equilibrium.

Answer:

$$m \cdot x_0 = (M + m)x_{cm}$$

We will find the coordinate of the center of mass by placing the MCN on the edge of the table so that the MCN almost turns over. Let's take 2 measurements. In the first case, we smoothly transfer the MCH from a vertical position with a weight from below to a horizontal one, in the second, from a vertical position with a weight on top. This is necessary to average the static friction force. Note. An alternative way to eliminate friction is to tap or roll the box on the table to bring the weight into equilibrium.

A3

0.60 pts

Experimentally determine the value of $m \cdot x_0$ and estimate its error.

Answer:

$$x_{cm1} = 2,1 \text{ cm}$$

$$x_{cm2} = 1,9 \text{ cm}$$

$$x_{cm} = \frac{x_{cm1} + x_{cm2}}{2} = 2,0 \text{ cm}$$

$$m \cdot x_0 = (400 \pm 20) \text{ g} \cdot \text{cm}$$

B0

0.10 pts

Measure and record the exact masses of the anchors and pancake given to you.

Answer:

$$m_0(100 \text{ g}) = 100.75 \text{ g}$$

$$m_{\text{ball}} = 99.89 \text{ g}$$

$$m_0(\text{disk}) = 40.78 \text{ g}$$

B1

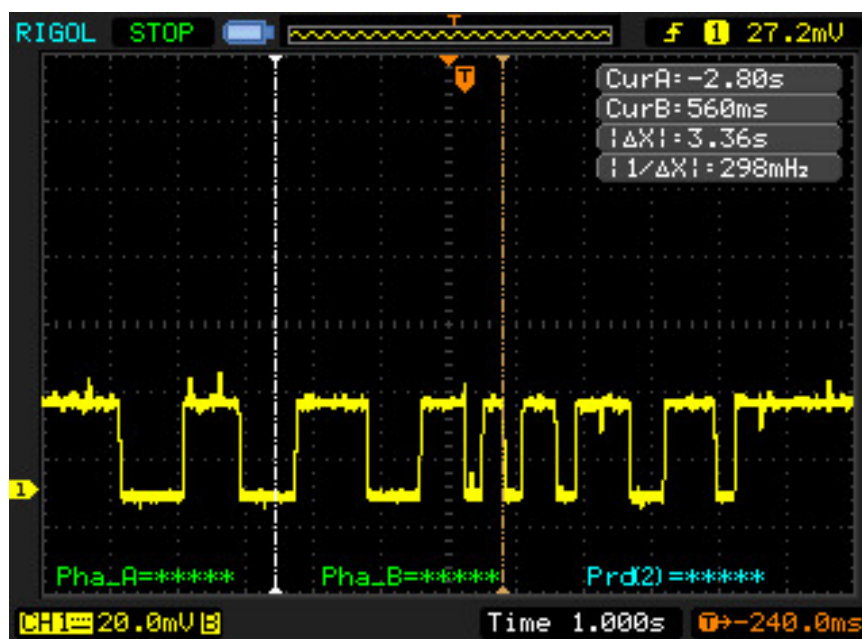
0.60 pts

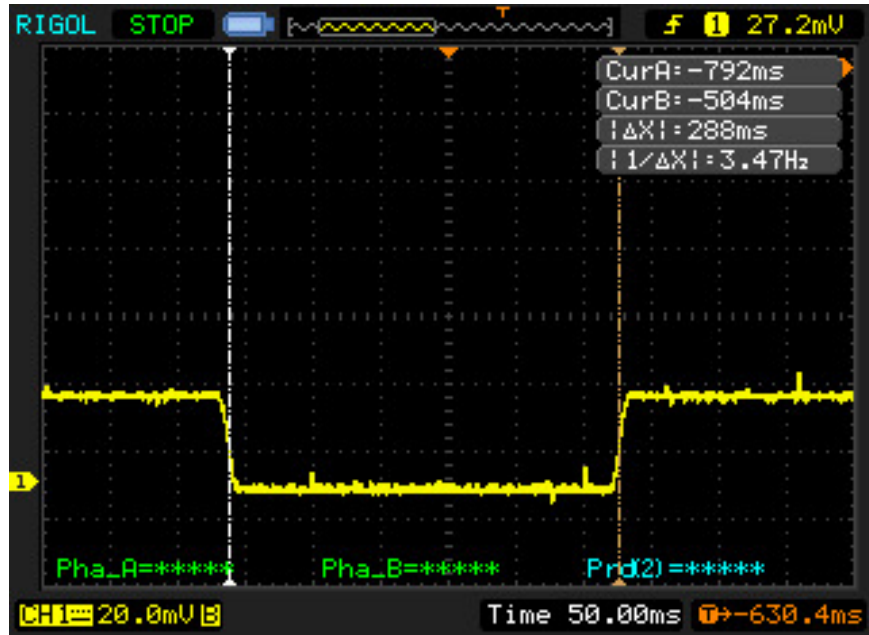
Sketch the characteristic appearance of the signal on an oscilloscope when the armature passes through the photogate. Specify between which points you will measure the time of passage of the anchor.

Typical appearance of an oscilloscope signal.

Typical view of the oscilloscope signal (approximate).

Answer: We will measure the time between the midpoints of the inclined sections (the cursors are positioned at the measurement points). Because the weight is large enough, it will gradually cover the space between the LED and the photodiode, and at the moment when the intensity drops by 2 times, the weight will cover half of the space, i.e. the lower (upper) edge of the weight will be located just between the LED and the photodiode.





B2

2.00 pts

Measure the speed of the armature v for various values of its downward displacement h in the range $[5; 90]$ cm in steps of 5 cm. For each h , measure the speed at least 3 times. Attention! After each change in the position of the photogates, make sure that the anchor does not touch them when moving!

Attention! After each change in the position of the photogates, make sure that the anchor does not touch them when moving!

$$d = 3.9 \text{ cm}$$

$h, \text{ mm}$	$\tau_1, \text{ ms}$	$\tau_2, \text{ ms}$	$\tau_3, \text{ ms}$	$\tau, \text{ ms}$	$v, \text{ m/s}$	$v^2, \text{ m}^2/\text{s}^2$	30	560	576	564	567	0.069	0.0047	40	556	574	578				
569	0.069	0.0047	60	426	436	438	433	0.090	0.0081	60	420	412	416	416	0.094	0.0088	90	348	358	342	349
0.112	0.0125	100	316	312	324	317	0.123	0.0151	120	310	306	306	307	0.127	0.0161	120	308	312	308	309	
0.126	0.0159	150	278	288	278	281	0.139	0.0192	180	264	264	268	265	0.147	0.0216	190	262	264	262	263	
0.148	0.0220	210	242	244	244	243	0.160	0.0257	240	234	234	234	234	0.167	0.0278	240	224	230	220	225	
0.174	0.0301	270	226	234	232	231	0.169	0.0286	300	218	212	214	215	0.182	0.0330	330	206	214	210	210	
0.186	0.0345	335	208	208	208	208	0.188	0.0352	360	214	208	212	211	0.185	0.0341	390	196	198	199	198	
0.197	0.0389	390	198	200	202	200	0.195	0.0380	420	197	196	194	196	0.199	0.0397	450	189	193	190	191	
0.205	0.0418	480	194	194	192	193	0.202	0.0407	485	182	184	178	181	0.215	0.0463	510	195	196	196	196	
0.199	0.0397	540	191	194	188	191	0.204	0.0417	540	176	188	186	183	0.213	0.0453	570	190	192	191	191	
0.204	0.0417	600	190	190	190	190	0.205	0.0421	630	187	190	189	189	0.207	0.0427	655	180	180	178	179	
0.217	0.0473	660	178	179	180	179	0.218	0.0475	690	178	181	177	179	0.218	0.0476	700	170	170	170	170	
0.229	0.0526	720	174	172	170	172	0.227	0.0514	750	172	171	170	171	0.228	0.0520	780	175	178	176	176	
0.221	0.0489	795	178	188	172	179	0.217	0.0473	810	175	176	177	176	0.222	0.0491	840	177	175	178	177	
0.221	0.0487	870	181	181	179	180	0.216	0.0468	900	177	179	178	178	0.219	0.0480	900	182	178	180	180	
0.217	0.0469																				

$$d = 3.9 \text{ cm}$$

B3

0.60 pts

Show that $v^2 = \gamma_1 h$ is in the slow rotation range and $v^2 = \gamma_2 h + b$ is in the fast rotation range.

Answer: The frictional force is equivalent to an increase in the moment of inertia of all rotating parts

by $\frac{1}{\alpha}$ times. Then we can write the ZSE:

$$\frac{I\omega^2}{2\alpha} + \frac{m_0 v^2}{2} + \text{const} = m_0 g h$$

thus. to. $v = \omega R$

$$v^2 + \text{const} = \gamma h$$

In the case of slow rotation $v(h=0) = 0$, means $\text{const} = 0$. Then can write for slow and fast rotation correspondingly:

$$v^2 = \gamma_1 h$$

$$v^2 = \gamma_2 h + b$$

B4

0.80 pts

Express the coefficients γ_1 and γ_2 in terms of $m, m_0, x_0, L, \delta, g, \alpha, R, I_B$ and I_1 .

$$m_0 \ddot{h} = m_0 g - T I \dot{\omega} = \alpha T R$$

Eliminating T and substituting $\dot{h} = \omega R$ obtain:

$$\ddot{h} = \frac{\alpha m_0 g R^2}{I + \alpha m_0 R^2}$$

Use following equality $\gamma = \frac{dv^2}{dh} = 2\ddot{h}$:

$$\gamma = \frac{2\alpha m_0 g R^2}{I + \alpha m_0 R^2}$$

Then substitute expression for moment inertia obtain:

$$\gamma = \frac{2\alpha m_0 g R^2}{\alpha m_0 R^2 + I_B + I_1 + m x^2}$$

For slow case $x = x_0$, for fast $x = \frac{L}{2} - \delta$:

Answer:

$$\gamma_1 = \frac{2\alpha m_0 g R^2}{\alpha m_0 R^2 + I_B + I_1 + m x_0^2}$$

$$\gamma_2 = \frac{2\alpha m_0 g R^2}{\alpha m_0 R^2 + I_B + I_1 + m(\frac{L}{2} - \delta)^2}$$

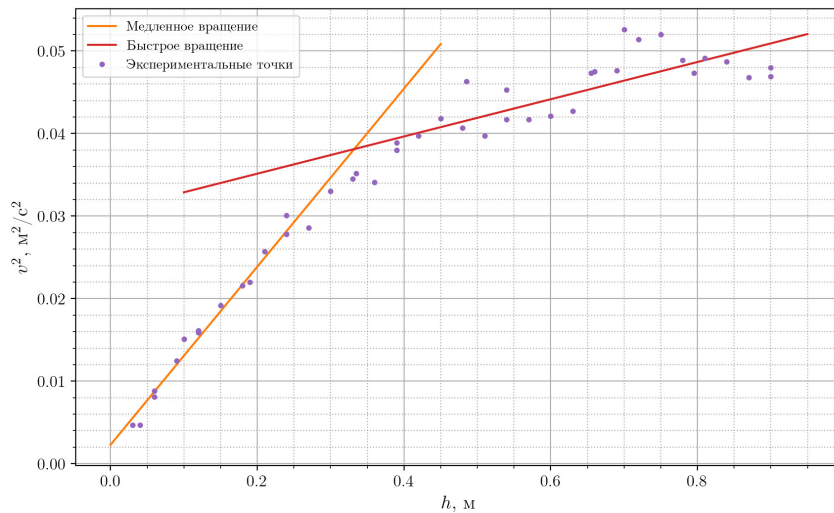
B5

1.20 pts

Construct a linearized graph of the relationship obtained in step B2. Indicate the ranges of slow and fast rotation on your answer sheets.

Characteristic appearance of the graph: Slow rotation at $v < 0.17 \frac{\text{m}}{\text{s}}$ Fast rotation at $v > 0.20 \frac{\text{m}}{\text{s}}$ In the section of the dependence where fast rotation is observed, oscillations with a characteristic period along h approximately 15 cm are visible. Note. Oscillations at the end of the graph arise from vibrations of the table, the characteristic period of which coincides with the perimeter of the cylindrical stand.

Note. Oscillations at the end of the graph arise from vibrations of the table, the characteristic period of which coincides with the perimeter of the cylindrical stand.



B6

2.00 pts

Take similar measurements for the new anchor. Change h in the range $h \in [5; 90]$ cm in increments of 5 cm. For each h , measure the speed at least 3 times.

$$d = 4.5 \text{ cm}$$

$h, \text{ mm}$	$\tau_1, \text{ ms}$	$\tau_2, \text{ ms}$	$\tau_3, \text{ ms}$	$\tau, \text{ ms}$	$v, \text{ m/s}$	$v^2, \text{ m}^2/\text{s}^2$
31	764	828	808	800	0.056	0.0032
50	936	944	941	941	0.048	0.0023
62	664	632	656	651	0.069	0.0048
75	736	720	744	733	0.061	0.0038
92	584	580	584	583	0.077	0.0060
100	636	648	648	644	0.070	0.0049
112	548	564	564	559	0.081	0.0065
125	584	572	568	575	0.078	0.0061
150	568	560	552	560	0.080	0.0065
150	516	492	512	507	0.089	0.0079
175	528	508	512	516	0.087	0.0076
180	472	472	460	468	0.096	0.0092
200	468	484	480	477	0.094	0.0089
211	436	448	448	444	0.101	0.0103
225	448	432	432	437	0.103	0.0106
245	412	404	408	408	0.110	0.0122
250	408	424	416	416	0.108	0.0117
280	390	400	392	394	0.114	0.0130
300	388	396	392	392	0.115	0.0132
336	380	380	392	384	0.117	0.0137
350	368	384	368	373	0.121	0.0145
400	348	352	352	351	0.128	0.0165
400	354	340	348	347	0.130	0.0168
446	348	334	352	345	0.131	0.0170
450	336	352	352	347	0.130	0.0169
500	332	328	328	329	0.137	0.0187
503	308	330	320	319	0.141	0.0199
535	320	316	310	315	0.143	0.0204
550	312	304	312	309	0.145	0.0212
585	314	308	311	311	0.145	0.0209
600	304	308	304	305	0.147	0.0217
650	310	306	300	305	0.147	0.0217
696	294	300	288	294	0.153	0.0234
700	298	298	296	297	0.151	0.0229
750	280	286	286	284	0.158	0.0251
790	280	286	282	283	0.159	0.0253
800	276	274	268	273	0.165	0.0272
850	280	276	274	277	0.163	0.0265
855	288	284	278	283	0.159	0.0252
900	276	270	272	273	0.165	0.0272

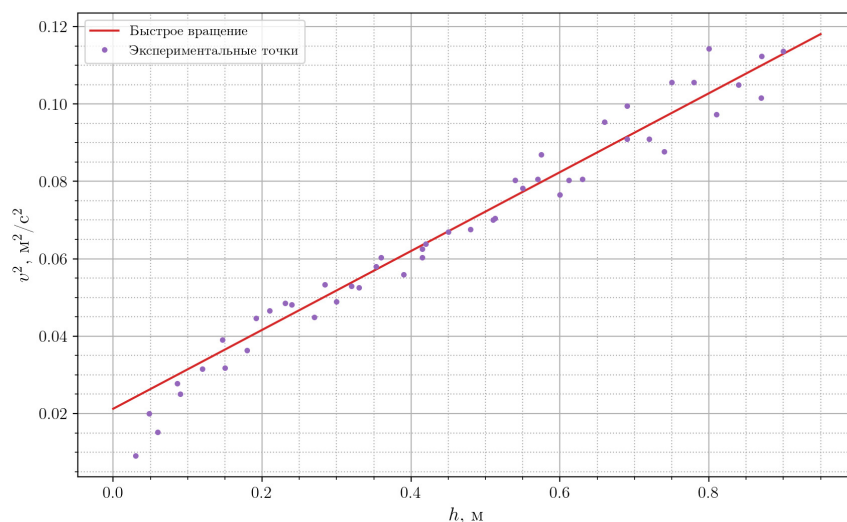
$$d = 4.5 \text{ cm}$$

B7

1.00 pts

Plot a linearized graph. In the answer sheets, highlight the type of rotation that is best explored using the obtained relationship.

The entire dependence is in the range of slow rotation.



B10

0.80 pts

Using the results from B4, B7, B9, obtain an expression for the mass m of the weight. Express your answer through m_{01} - the mass of the light anchor, m_{02} - the mass of the heavy anchor, L , δ , R , α , $m \cdot x_0$, g , γ_1 , γ_2 .

$$x_0 = \frac{(mx_0)}{m}.$$

Rewrite the equations in the following form:

$$\begin{cases} I_B + I_1 + m \left(\frac{L}{2} - \delta \right)^2 + \alpha m_{02} R^2 = 2\alpha R^2 g \frac{m_{02}}{\gamma_2}, \\ I_B + I_1 + mx_0^2 + \alpha m_{01} R^2 = 2\alpha R^2 g \frac{m_{01}}{\gamma_1}. \end{cases}$$

Subtracting gives the equation:

$$m \left(\left(\frac{L}{2} - \delta \right)^2 - \frac{(mx_0)^2}{m^2} \right) = 2\alpha R^2 \left(\frac{m_{02}g}{\gamma_2} - \frac{m_{01}g}{\gamma_1} + \frac{m_{01} - m_{02}}{2} \right)$$

Solving the quadratic equation we get the answer:

Answer:

$$m = \frac{\alpha R^2 \left(\frac{m_{02}g}{\gamma_2} - \frac{m_{01}g}{\gamma_1} + \frac{m_{01} - m_{02}}{2} \right) + \sqrt{\alpha^2 R^4 \left(\frac{m_{02}g}{\gamma_2} - \frac{m_{01}g}{\gamma_1} + \frac{m_{01} - m_{02}}{2} \right)^2 + \left(\frac{L}{2} - \delta \right)^2 (mx_0)^2}}{\left(\frac{L}{2} - \delta \right)^2}$$

B11

1.20 pts

Calculate the values of m and x_0 and estimate their errors.

Answer:

$$m = (80 \pm 30) \text{ g}$$

$$x_0 = (5 \pm 2) \text{ cm}$$

C0

0.10 pts

Measure and record the distance D .

Answer:

$$D = (3.7 \pm 0.1) \text{ cm}$$

C1

0.50 pts

Write down an expression for the effective stiffness k of such a spring system in terms of k_1 and k_2 .

Let's consider the forces acting on the weight from the springs (axis x is directed towards k_1):

$$F = k_1 \left(\frac{L}{2} - x \right) - k_2 \left(\frac{L}{2} + x \right)$$

$$F = \frac{L}{2} (k_1 - k_2) - x(k_1 + k_2)$$

It can be seen, that for a displacement, on the weight there acts a force $\Delta F = (k_1 + k_2)\Delta x = k\Delta x$

Answer:

$$k = k_1 + k_2$$

C2

1.20 pts

Measure the periods T_1 of oscillation of the MCH as a pendulum in the first position (in the figure on the left) and T_2 in the second position (in the figure on the right). Attention! Take measurements with the greatest possible accuracy, because... this is important to obtain an accurate result.

Attention! Take measurements with the greatest possible accuracy, because... this is important to obtain an accurate result.

Let's measure the time of 150 oscillations 2 times for each position: $t_{11} = 189.08$ s $t_{12} = 189.35$ s $t_{21} = 182.96$ s $t_{22} = 183.10$ s Then the periods:

$$t_{11} = 189.08 \text{ s } t_{12} = 189.35 \text{ s } t_{21} = 182.96 \text{ s } t_{22} = 183.10 \text{ s}$$

Then the periods are:

$$T_2 = (1.220 \pm 0.001) \text{ s}$$

Answer:

$$T_1 = (1.261 \pm 0.001) \text{ s}$$

$$T_2 = (1.220 \pm 0.001) \text{ s}$$

C3

0.80 pts

Express the periods of oscillation T_1 and T_2 in terms of M , m , L , D , x_0 , Δx , g , I_0 and I_1 .

Let us write down the moments of forces acting on the pendulum:

$$I\ddot{\varphi} = -\varphi \left(Mg \left(\frac{L}{2} + D \right) + mg \left(\frac{L}{2} + D \pm x_0 + \Delta x \right) \right)$$

The sign is chosen in accordance with the orientation of the MCH. Then the periods of oscillation:

Answer:

$$T_1 = 2\pi \sqrt{\frac{I_0 + I_1 + m \left(\frac{L}{2} + D + x_0 + \Delta x \right)^2}{Mg \left(\frac{L}{2} + D \right) + mg \left(\frac{L}{2} + D + x_0 + \Delta x \right)}}$$

$$T_2 = 2\pi \sqrt{\frac{I_0 + I_1 + m \left(\frac{L}{2} + D - x_0 + \Delta x \right)^2}{Mg \left(\frac{L}{2} + D \right) + mg \left(\frac{L}{2} + D - x_0 + \Delta x \right)}}$$

C4

1.40 pts

Excluding the moment of inertia I_0 from the expressions obtained in the previous paragraph, obtain an expression for Δx . Calculate the numerical value of Δx and estimate the error.

Let's rewrite the equations in the following form:

$$\begin{cases} I_0 + I_1 + m \left(\frac{L}{2} + D + x_0 + \Delta x \right)^2 = \frac{T_1^2}{4\pi^2} \left(Mg \left(\frac{L}{2} + D \right) + mg \left(\frac{L}{2} + D + x_0 + \Delta x \right) \right) \\ I_0 + I_1 + m \left(\frac{L}{2} + D - x_0 + \Delta x \right)^2 = \frac{T_2^2}{4\pi^2} \left(Mg \left(\frac{L}{2} + D \right) + mg \left(\frac{L}{2} + D - x_0 + \Delta x \right) \right) \end{cases}$$

Subtract these equations and express Δx :

Answer:

$$\Delta x = \frac{\frac{T_1^2 - T_2^2}{4\pi^2} (M + m) \left(\frac{L}{2} + D \right) g + \frac{T_1^2 + T_2^2}{4\pi^2} mg x_0 - 2m x_0 (L + 2D)}{4m x_0 - \frac{T_1^2 - T_2^2}{4\pi^2} mg}$$

For calculation we must take into account that the mass M has changed because of the attachment of the nuts with the loop on $2M_g = 94.22 \text{ g}$

$$\Delta x = (3.2 \pm 1.9) \text{ cm}$$

C5

0.20 pts

Write an expression for the effective spring stiffness k in terms of m , g , Δx . Calculate the numerical value of k and estimate the error.

Answer:

$$k = \frac{mg}{\Delta x}$$

$$k = (24 \pm 18) \frac{\text{N}}{\text{m}}$$

C6

0.80 pts

Using the results from Parts B and C, calculate the values of k_1 , k_2 and estimate their errors.

From C1:

$$F(x = x_0) = 0 = \frac{L}{2} (2k_1 - k) - kx_0$$

Express k_1 :

$$k_1 = k \left(\frac{1}{2} + \frac{x_0}{L} \right) = k \left(\frac{1}{2} + \frac{x_0}{L} \right)$$

$$k_2 = k \left(\frac{1}{2} - \frac{x_0}{L} \right)$$

$$F(x = x_0) = 0 = \frac{L}{2} (2k_1 - k) - kx_0$$

Expressing k_1 :

$$k_1 = k \left(\frac{1}{2} + \frac{x_0}{L} \right) = k \left(\frac{1}{2} + \frac{x_0}{L} \right)$$

$$k_2 = k \left(\frac{1}{2} - \frac{x_0}{L} \right)$$

$$k_1 = (7 \pm 4) \frac{\text{N}}{\text{m}}$$

Answer:

$$k_1 = (17 \pm 13) \frac{\text{N}}{\text{m}}$$

$$k_1 = (7 \pm 4) \frac{\text{N}}{\text{m}}$$